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Professional Summary

An AI/LLM Scientist with a **Ph.D. in Computer Science** specializing in **NLP and Large Language Models**, and **2+ years of industry** experience at Amazon building Agentic systems for **millions of Amazon customers**. Expert in **model training/evaluation, agentic systems, context management, RAG and personalization**. Author of **20+ publications (8 first-authored)** in top venues (NeurIPS, ACL, EMNLP, NAACL etc.) and recipient of a **US Patent**. Lead author of **pioneering works in early multi-agent systems and hyper-attention based tabular LLMs**.

On the [Amazon Rufus](#) (Shopping Assistant) **foundation modeling team**, works on core model capabilities for customer-facing agentic systems at scale, including extending the context window from **8K to 128K** through continual pre-training, and improving faithfulness, personalization, and multi-turn context handling through **DPO/GRPO-based** post-training. In **Amazon Customer Service**, leads the end-to-end LLM **data flywheel** for the **multi-agent system**, spanning live traffic failure monitoring, automated root-cause analysis, training and evaluation data synthesis, and model deployment, enabling repeatable closed-loop model improvement with impact across workstreams.

Experience

Amazon <i>Applied Scientist (full-time)</i>	2024.1 – Present Seattle, WA & Palo Alto/Santa Clara, CA
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Led model training research and development for customer-facing LLM agent systems, focusing on context management, personalization, response faithfulness and closed-loop data flywheel.

- **At Amazon Rufus Foundation Modeling Team:** Co-led the context management and RAG workstream,
 - reduced hallucinations and improved model faithfulness from **67% to 90%** through synthetic IFT/DPO data;
 - improved personalization helpfulness from **84% to 93%** with verifiable rubrics and GRPO;
 - extended the context window from **8K to 128K** through continual pre-training with ABF and YaRN;
 - and increased multi-turn context carryover from **61% to 73%** using DPO with live-traffic failure patterns as negatives.
- **At Amazon Customer Service Team:** Owned the end-to-end LLM **data flywheel for the team's multi-agent system**, spanning live-traffic failure monitoring, automated root-cause analysis, post-training data generation and verification, and model training/evaluation. Established automated quality gates and repeatable training → measurement loops across workstreams, e.g.,
 - diversity-aware data selection improved agent tool selection accuracy from **86% to 93%**, while reducing training time by **4x (24h→6h)**.
 - difficulty-aware data selection improved response generation quality from **88% to 91%**, while reducing training time by **5x (10d→2d)**.
- **Intern Mentorship:** Mentored two interns on LLM research projects in long-context modeling, and multi-turn improvement with verifiable GRPO, guiding ideation, experimentation, evaluation, and writing, resulting in **two published paper** and one paper under review.

Amazon Web Service (AWS) <i>Applied Scientist (Intern)</i>	2023.5 – 2023.8 Santa Clara, CA
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- Researched LLM-based multi-agent systems for complex problem solving.
- Developed a **collaborative multi-agent** prompting framework that achieved state-of-the-art performance on complex science benchmarks in both zero-shot and few-shot settings.
- Published the [work](#) in Findings of NAACL 2024.

Amazon Web Service (AWS)	2022.6 – 2023.1
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Applied Scientist (Intern)

Santa Clara, CA & remote

- Developed a novel **tabular language modeling** approach that represents tables as hypergraphs to better capture structural dependencies.
- Achieved state-of-the-art performance on tabular representation learning benchmarks.
- Published the [work](#) at NeurIPS 2023 as a Spotlight paper (**top 5%**).

Tencent AI Lab

2021.6 – 2021.8

NLP Researcher (Intern)

remote in College Station, TX

- Built a comprehensive benchmark for zero-shot knowledge base completion, covering state-of-the-art methods and diverse knowledge sources.
- Resulted in a [paper](#) accepted to the ICDM 2022 Knowledge Graph Workshop.

Texas A&M University

2019.9 – 2023.12

Research Assistant and Teaching Assistant

College Station, TX

- Conducted research in NLP and information extraction, including named entity recognition, event extraction, and fine-grained opinion mining.
- Developed graph-based methods to model non-sequential dependencies among entity mentions, improving domain-specific NER and structured information extraction.
- Built datasets and benchmarks for fine-grained language understanding tasks, including location recognition.
- Served as a teaching assistant for CSCE 636 Deep Learning and CSCE 313 Computer Systems.

National Lab of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences

2018.1 – 2019.7

Research Engineer (Intern & Full-time)

Beijing, China

- Built information extraction and evaluation systems for financial event extraction and causality detection, bridging NLP research and real-world financial text applications.

Innovation Lab of Global Exchange, State Street

2017.7 – 2018.1

Data Analyst (Intern)

Hangzhou, China

- Supported data cleaning, analysis, visualization, and database construction for financial applications.

Selected Publications

LLM & Foundation Model Training

- **Pei Chen**, Soumajyoti Sarkar, Leonard Lausen, et al. "[HYTREL: Hypergraph-enhanced Tabular Data Representation Learning](#)." **NeurIPS 2023, Spotlight (5%), first work to use hyper-attention for tabular LLMs.**
- Ming Li, **Pei Chen**, Chenguang Wang, et al. "[Mosaic-IT: Cost-Free Compositional Data Synthesis for Instruction Tuning](#)." ACL 2025, Findings.
- Hongye Jin, **Pei Chen**, Jingfeng Yang, et al. "[END: Early Noise Dropping for Efficient and Effective Context Denoising](#)." pre-print.
- Wangtao Sun, Haotian Xu, Xuanqing Yu, **Pei Chen**, et al. "[ItD: Large Language Models Can Teach Themselves Induction through Deduction](#)." ACL-2024, long paper.
- Zhaoxuan Tan, Zheng Li, Tianyi Liu, Haodong Wang, **Pei Chen**, et al. "[Aligning Large Language Models with Implicit Preferences from User-Generated Content](#)." ACL 2025, long paper.

Agent

- **Pei Chen**, Shuai Zhang, Boran Han. "[CoMM: Collaborative Multi-Agent, Multi-Reasoning-Path Prompting for Complex Problem Solving](#)." NAACL 2024, long paper, **pioneering work in demonstrating multi-agent systems for complex LLM reasoning.**
- Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, Xin Liu, **Pei Chen**, et al. "[Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training](#)." NAACL 2025, long paper.
- Hy Dang, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, **Pei Chen**, et al. "[Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates](#)." EMNLP 2025, long paper.
- Ziyi Wang, Yuxuan Lu, Yimeng Zhang, **Pei Chen**, et al. "[Trajectory2Task: Training Robust Tool-Calling Agents](#)

with Synthesized Yet Verifiable Data for Complex User Intents." accepted to ACL 2026, long paper.

RAG & Long-context & Multi-turn & Personalization

- **Pei Chen**, Hongye Jin, Cheng-Che Lee, et al. "LongLeader: A Comprehensive Leaderboard for Large Language Models in Long-context Scenarios." NAACL 2025, long paper.
- Ming Li, **Pei Chen**, Zhenhao Zhang, et al. "Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards." Accepted to ACL 2026, long paper.
- Yichuan Li, Xinyang Zhang, Chenwei Zhang, Mao Li, **Pei Chen**, et al. "ALERT: An LLM-powered Benchmark for Automatic Evaluation of Recommendation Explanations." NAACL 2025, long paper.
- Fengran Mo, Yifan Gao, Chuan Meng, Xin Liu, **Pei Chen**, et al. "UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations." ACL 2025, long paper.
- Fengran Mo, Yifan Gao, Zhuofeng Wu, Xin Liu, **Pei Chen**, et al. "Leveraging historical information to boost retrieval-augmented generation in conversations." Information Processing & Management, March 2026.
- Zhaoxuan Tan, Zixuan Zhang, Haoyang Wen, Zheng Li, **Pei Chen**, et al. "Instant Personalized Large Language Model Adaptation via Hypernetwork." , accepted to ACL 2026.
- Yizhuo Chen, Xin Liu, Ruijie Wang, Zheng Li, **Pei Chen**, et al. "POPI: Personalizing LLMs via Optimized Natural Language Preference Inference." pre-print.

Information Extraction

- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. "Explicitly Capturing Relations between Entity Mentions via Graph Neural Networks for Domain-specific Named Entity Recognition." ACL-IJCNLP 2021, short paper.
- **Pei Chen**, Haotian Xu, Cheng Zhang, and Ruihong Huang. "Crossroads, Buildings and Neighborhoods: a Dataset for Fine-grained Location Recognition." NAACL 2022, long paper.
- **Pei Chen**, Hang Yang, Kang Liu, et al. "Reconstructing Event Regions for Event Extraction via Graph Attention Networks." AACL-IJCNLP 2020, long paper.
- **Pei Chen**, Haibo Ding, Jun Araki, and Ruihong Huang. "Domain-specific Named Entity Recognition via Graph Neural Networks." US Patent 12,530,529, 2026.

Education

Texas A&M University	2019.8 - 2024.5
<i>Ph.D. in Computer Science</i>	<i>College Station, TX</i>

- **GPA:** 4.0/4.0
- **Research Directions:** Natural language processing, with a focus on information extraction and structured language understanding, including named entity recognition, event extraction, and knowledge base completion; later extended to large language models and agentic systems through research and applied internships.
- **Dissertation:** Connecting the Dots: Improving Information Extraction by Modeling the Non-Sequential Dependencies of Entity Mentions.

Southwestern University of Finance and Economics	2016.9 – 2018.6
<i>Master's Degree in Finance</i>	<i>Chengdu, China</i>

- **GPA:** 3.9/5.0, ranking 1/178.
- **Thesis:** Does News Sentiment Predict the Stock Market? An Example on Chinese Growth Market.
- **Awards:** 2017 National Scholarship for Graduate Student.

National University of Defense Technology	2010.9 – 2014.6
<i>B.Eng. in Simulation Engineering</i>	<i>Changsha, China</i>

- **GPA:** 88.61/100, ranking 1/45.
- **Thesis:** Analyze and reconstruct the multi-resolution modeling technology of a simulation system written by millions of lines of C++ code.

Professional Service

- 2026: **Area Chair for ARR**, Program Committee/Reviewer for ARR, ICLR 2026, ICML 2026.
- 2025: Program Committee/Reviewer for ARR, COLING 2025, ICLR-2025, NeurIPS 2025
- 2024: Program Committee/Reviewer for ARR, NeurIPS 2024, TPAMI
- 2023: Program Committee/Reviewer for ACL, EMNLP, ARR
- 2022: Program Committee/Reviewer for EMNLP, ARR, NLPCC
- 2021: Program Committee/Reviewer for EMNLP, ARR, NLPCC

Skills

- **LLM & ML:** Distributed Training, Post-training, Supervised Fine-tuning, Preference Optimization, DPO, GRPO, DAPO, Context Management, Personalization, Offline/Online Evaluation, Rubric Design, Long-Context Modeling, Synthetic Data, Multi-Agent Systems Design, RAG, Hallucination Mitigation, Data Flywheel
- **Programming & Frameworks:** Python, PyTorch, Git, Docker, SQL, C/C++ , Weights & Biases